Figure and table legends

Table 1. Attrition funnel of the development draw, showing the count surviving each pipeline stage, its share of the attempted pairs and of the preceding stage, and the dominant rejection reason at that stage.

Table 2. Error profile of the raw midpoint, the bound-filtered midpoint, the identified-but-uncertified set and the certified set against the benchmark's defined truth, on each method's own set.

Table 3. The selection frontier: for each certification tolerance, the size and coverage of the certified set, its mean absolute error, the error it abstains from, and the pair-null and day-block-null means with their add-one p-values.

Table 4. The distribution, over the 23 replication draws, of certified coverage, certified error and raw-midpoint error, reported as median, interquartile range and full range.

Table 5. The pre-registered rule tally over the 23 replication draws, reporting each of the six rules as passed or failed.

Table 6. Certified coverage and certified and raw-midpoint error across the ten market specifications of the sweep, at the primary tolerance.

Table 7. Response of certified coverage, error and mean absolute CPIV change to each of the six declared perturbations, on the development draw.

Figure 1. Two panels: (a) the certified mean absolute error against the certification tolerance, with the pair-null 5th to 95th percentile band and the day-block null mean; (b) certified coverage across the ten market specifications of the sweep.

Two further tables are supplied as supplementary material: Table S1, the strata of certified coverage by moneyness, maturity, spread and true magnitude, and Table S2, the declared settings of the protocol.

Abstract

Call–put implied-volatility spreads (CPIV) computed from quote midpoints are routinely reported as measurements; a midpoint says nothing about whether the spread is determined by the quotes or by their placement within the bid–ask spread. This paper describes an abstention-aware protocol that either certifies a CPIV as identified at a declared tolerance or abstains, giving every attempted pair exactly one terminal status so nothing is silently dropped. The protocol does not re-estimate the spread: where it and the raw midpoint are both defined they are numerically identical, so every own-set improvement is a selection effect. The abstention rule is tested against size-matched random subsets, under a pair null and a day-block null that preserves within-day dependence. With no licensed quote data available, the protocol is validated on a synthetic instrument whose true spread is defined before any quote exists, under a pre-registration frozen beforehand. In the benchmark the certified subset beats size-matched random subsets in every draw. One pre-registered rule fails: certification coverage depends strongly on quote microstructure, falling towards zero on coarse markets. It is a labelling device whose coverage is a property of the market it meets, not a rule with a fixed yield.

Keywords

call–put implied-volatility spread; option-quote measurement; abstention-aware inference; identification; Black–Scholes–Merton inversion; numerical validation

Key messages

- Certified coverage falls towards zero on coarse-quote markets
- Abstention selects a genuinely easier set, not merely a smaller one
- On the common set the protocol is identical to the raw midpoint
- Thresholds were frozen before any replication output was read

When Is a Call–Put Implied-Volatility Spread Measurable? An Abstention-Aware Protocol Validated Against a Known Truth

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1. Introduction

1.1 A number that is not always a measurement

The call–put implied-volatility spread is a natural object to compute from option market data. For a call and a put on the same underlying with the same strike and expiry, the two implied volatilities are inverted from their own quotes under a common pricing map and differenced. The spread is used as a feature in models of option mispricing (Cremers & Weinbaum, 2010) and as a diagnostic of the shape of the volatility surface, and in each use it is treated as a *quantity* — a number that exists in the data, to be read off and compared across securities and dates. That treatment is a choice, and not a harmless one. A quote is an interval: a bid and an ask, separated by a spread that varies with liquidity, moneyness and maturity (Madhavan, 2000), and carrying a timestamp, and the midpoint of that interval is a convention, not an observation. That the spread carries information about informed trading, not merely a liquidity cost, is the dealer-quote literature's central result (Glosten & Milgrom, 1985); the midpoint discards it. Inverting a midpoint and differencing two such numbers produces numbers, but neither step says whether the result is determined by the market or by where the quotes happened to sit; the error is governed by the width of the two spreads, not by the surface the analyst believes they are measuring. The consequence is a common failure. A CPIV table is produced, its cross-sectional dispersion is studied, and the dispersion is read as a property of the market. Some of it is. Some of it is the quote microstructure, amplified by the inversion and re-expressed in volatility units. The two are not separable afterwards, because the information needed to separate them was discarded when the midpoint was taken.

1.2 What the standard toolkit does and does not do

The measurement literature has a well-developed toolkit for this problem, and the protocol described here draws on it rather than replacing it. A bound filter removes pairs whose quotes violate the no-arbitrage relations that option prices must satisfy (Merton, 1973); such violations are common in dealer quotes and are informative about the limits to arbitrage rather than being mere data errors (Figlewski, 1989). A bound-filtered midpoint replaces the raw midpoint. An endpoint interval reports the range of CPIV implied by the four bid–ask

endpoint combinations, which is a strictly more honest object than a point. A parity residual cross-checks the algebra, though one that is algebraically dependent on the CPIV itself and therefore not an independent validation. Parity deviations are also studied in their own right, as evidence of short-sale constraints and segmented markets (Ofek et al., 2004). Each is a genuine improvement on the raw midpoint, and each is reported here as a declared baseline. What none of them does is decline to answer. A bound-filtered midpoint is still a point estimate of a quantity the quotes may not determine. An endpoint interval is honest about its own width, but the width is usually not carried forward. A parity residual checks the algebra, not the identification of the quantity.

1.3 The central claim, and its awkward consequence

The protocol produces a CPIV by taking the midpoint of the same interval the raw baseline takes the midpoint of. It does not improve the estimator. It labels pairs.

That has a consequence that is easy to state and easy to avoid stating: on the set of pairs where the protocol certifies a value and the raw midpoint also produces one, the two values are *numerically identical*. Every difference in error between the two methods on that set is exactly zero, by construction, and not because the two methods happen to agree.

It follows that any apparent improvement the protocol delivers is delivered by *selection*: by declining to report certain pairs. A reported subset of the pairs that are actually identified is more useful than a reported set that silently includes pairs that are not, but an improvement delivered by selection is not the same kind of claim as an improvement delivered by a better estimator, and a reader is entitled to know which is on offer: an effect estimated on the treated group is not an estimate of the effect of treatment, and the comparison must be made against the right counterfactual (Imbens & Rubin, 2015).

So the protocol's own-set advantage is stated as what it is — a selection effect — and the substantive question becomes: *is the certified subset genuinely easier to measure than an arbitrary subset of the same size?* If it is, the abstention rule is doing something real. If it is not, the protocol is discarding hard pairs and paying for the privilege in coverage. Answering that requires a counterfactual that respects the dependence in the data, and this paper constructs two size-matched nulls for each tolerance.

1.4 Why the validation is synthetic

The obvious way to answer the previous question is with real quotes. That is not currently possible, and the reason is not a matter of taste or effort. No licensed product covering the required data was available at the time of writing, and the data gate for this project is deliberately closed: no market-data path will run without a written licence record permitting publication of the aggregate outputs, and the enquiry letters to the two candidate vendors have been sent, with no licence granted yet. Options market data is not freely redistributable, and tables derived from quote data one may not redistribute cannot be published. The alternative taken here is to make the *method* the object of validation and to give the method a test a real market could not have provided. A synthetic market can be constructed in which the true call–put implied volatility spread is defined *before* any quote exists. The generator draws fair values, inverts them under the same map the protocol uses, and thereby knows the truth; only then does it round to a tick, add a spread, jitter the timestamps and inject defects. In a real market the protocol can be assessed for coverage and stability, but never for accuracy, because the truth is unknown; in the benchmark it can be assessed for all three. This is not a substitute for the licensed run, and the paper does not present it as one. It supports a narrower

claim: that the measurement machinery behaves as specified. Section 7.2 states the non-claims, and they bind the abstract and the conclusion.

1.5 Contributions

1. **An abstention-aware measurement protocol with a closed status taxonomy.** Declining to answer a question the data cannot support, and reporting that refusal, is the optimum reject rule of the pattern-recognition literature (Chow, 1970) applied to measurement. Every attempted quote pair receives exactly one terminal status, the counts are reconciled against the attempted total, and a residual fails the run. Nothing can be dropped silently.
2. **A size-matched null test for an abstention rule.** A pair null drawn uniformly over the scored universe and a day-block null that preserves the within-day dependence the study's own intervals respect; the block null is the conservative test and is reported as primary.
3. **A discipline for reporting the protocol's own advantage.** The paper reports the identity that the protocol and the raw midpoint coincide on the common set, and requires every own-set comparison to carry the frontier p-values that say whether the selected set beats an arbitrary set of the same size.
4. **A pre-registered synthetic ground-truth benchmark with a rule tally in which one rule fails.** The generator, settings, seeds, specification grid, null construction and decision rules were fixed in a pre-registration addendum written before the replication sweep was run, and the tally reports which rules passed and which did not, with the same prominence.

2. The measurement problem

2.1 What a CPIV is made of

Let C and P denote a call and a put on the same underlying, with the same strike K, expiry T, exercise style and contract terms, quoted at a common time. Write σ_C and σ_P for the implied volatilities recovered from the two quotes under a common pricing map, and CPIV = $100 \times (\sigma_C - \sigma_P)$ in percentage points of volatility. For a European pair under the Black–Scholes–Merton map (Black & Scholes, 1973), both legs are determined by the same five inputs — spot, strike, maturity, rate and dividend yield — so σ_C and σ_P are two noisy readings of the same object, and their difference measures the *inconsistency* of the two quotes with the model. Two features matter for what follows. First, the difference is a difference of inverted quantities, so an error in either leg is amplified by the inverse of that leg's vega. Second, because both legs are inverted from intervals rather than points, the CPIV inherits the width of both quote intervals. Write the two half-spreads in price units as h_C and h_P and the two vegas as v_C and v_P; to first order the CPIV is uncertain by an amount of the order of $100 \times (h_C / v_C + h_P / v_P)$. A wing option, with a small vega and a wide spread, contributes a large and roughly constant uncertainty to the CPIV regardless of the true value. This is the mechanism behind most of the dispersion in a CPIV table, and it is why that dispersion is not interpretable as a market quantity.

2.2 Identification is not estimation

Two activities are kept separate throughout. *Estimation* is recovering a value well, given that a value exists. *Identification* is establishing that a value exists at all, given the data. The raw midpoint pipeline performs an estimation and silently assumes identification. A midpoint of a bid and an ask, inverted and differenced, is a well-defined number; nothing in the operation

establishes that the number is close to anything. The two can come apart badly. Consider a deep out-of-the-money put whose time value is a few ticks wide but whose vega is tiny. The midpoint of that put's interval can be inverted to a volatility that is off by a large relative amount, because the inversion divides a small price error by a small vega. The resulting CPIV is dominated by the error of the leg that cannot be measured, and it is reported with the same confidence as every other number in the table. The classical response is to declare in advance what would count as a successful measurement, and to report a null result where the criterion is not met. That is the structure adopted here: the protocol answers "is this pair's CPIV identified at tolerance t?" for each pair, in a way a downstream user cannot mistake for a measurement.

2.3 The two objects a protocol could produce

A protocol can produce a *second estimator*: a different way of computing the CPIV, expected to be closer to the truth, whose common-set difference would be non-trivial and whose fall in common-set error would be evidence about the estimator. Or it can produce a *labelling device*: a rule that selects a subset and declines to speak about the rest, whose output on the selected subset is whatever the original estimator would have produced there. The common-set difference is exactly zero, and any advantage is a function of which pairs were selected. The protocol here is of the second kind. The problem is not "the midpoint is the wrong number" but "the midpoint is a number about which nothing can be asserted without knowing whether it is determined". Its advantage must therefore be tested against a counterfactual, because the common-set comparison is vacuous by construction (Section 2.5).

2.4 Declared baselines and the comparison that matters

Five baselines are declared in the pre-registration, and each is reported: the raw midpoint CPIV, with no filter beyond the existence of a finite root; the bound-filtered midpoint CPIV; the bid–ask endpoint interval, reported as an interval rather than a point; the parity residual, labelled throughout as algebraically dependent on the CPIV and so not an independent validation; and the abstention-aware protocol itself, with its certified subset, its abstentions and its funnel. The comparison the paper is built around is between the fifth and the first, reported in two forms that must not be confused. On the **own set**, each method is evaluated on the pairs it can use: the protocol's own set is small and accurate, the raw midpoint's is large and inaccurate, and the difference is the selection effect. On the **common set**, both are evaluated on the pairs they can both use, the values are identical, and the difference is exactly zero — reported as exactly zero rather than as a small number.

2.5 The counterfactual: size-matched random subsets

Suppose the protocol certifies k of the n scored pairs. Its own-set mean absolute error is the mean absolute error of those k pairs. The natural null is the distribution of that statistic over subsets of size k drawn at random from the same n pairs. Two nulls are constructed, and the difference between them is the point. The **pair null** draws k pairs uniformly without replacement from the scored universe. It asks whether the certified set is better than a random selection of the same size, if selections were independent of everything. It is the natural null, and the one most likely to favour the protocol, because it ignores that pairs from the same underlying-day are not independent. The **day-block null** draws whole underlying-days with replacement until k pairs have been selected, so every draw respects the within-day dependence of the quotes. It asks the harder question: is the certified set better than a random selection that respects the same dependence structure the study's own confidence intervals respect? A protocol that beats a pair null but not a day-block null has selected a set whose

advantage comes from its correlation with the day's overall quote quality rather than from its own per-pair criteria. Both nulls are Monte Carlo with a declared seed and a declared number of trials. The null distribution is a bootstrap distribution in the sense of Efron and Tibshirani (1993), the day-block draw is a block resampling scheme for dependent data of the kind developed by Politis and Romano (1994), and the reported p-value is the add-one estimator (Davison & Hinkley, 1997). The day-block null is the primary test, because it is the conservative one and because the block resampling is the same device used for the study's own confidence intervals. One implementation detail matters, and is stated because it affects the result. A block draw must match the certified subset exactly in size, so the final block of a draw may be taken in part, its members chosen at random. Taking them systematically — always the first *m* members, say — would bias the null towards the earliest pairs of a day, which are not exchangeable with the rest, and would make the null artificially easy to beat.

2.6 What the microstructure does to the coverage

The protocol's coverage — the share of attempted pairs it certifies — is a property of the protocol *and* of the market it meets. Two rules control it directly. The resolution scale requires the option's out-of-the-money- equivalent time value to exceed a declared multiple of the tick before the pair is treated as resolvable at all; the tolerance requires the endpoint interval of the CPIV to be narrower than the declared tolerance. Both interact with microstructure in a way that is not a matter of degree. If the tick is coarse relative to the time value of a wing option, the resolution rule abstains on that option whatever the tolerance. If the quotes are wide, the endpoint interval is wide, and the tolerance rule abstains on nearly everything. A protocol that certifies a small, well-identified fraction of a liquid market may certify almost nothing of an illiquid or coarsely quoted one, and a reader who carries a coverage figure from one market to another will be wrong.

3. The protocol

3.1 Pair construction

An observation is a matched call–put pair. Two quotes are a pair only when they agree on underlying, strike, expiry, multiplier, exercise style and settlement convention; a disagreement in any of these fields is recorded as a contract mismatch, with the offending field named, and the pair is excluded from measurement without being dropped from the denominator. Each quote carries a timestamp, and each pair carries three lags: the call–put lag, the call–spot lag and the put–spot lag. A pair whose lags exceed the declared synchronisation tolerance is a timing mismatch, and the lags are retained on the record so the exclusion can be audited rather than trusted. The spot used for inversion is the underlying's own quote at its own timestamp, not a last-traded price or a value carried forward. This rules out end-of-day files, which give every option the same snapshot and make the timing stage vacuous.

3.2 The inversion contract

Both legs are inverted under the same European Black–Scholes–Merton map (Black & Scholes, 1973; Merton, 1973), whose state-contingent foundation is due to Breeden and Litzenberger (1978), with a single declared contract:

- the root bracket is decimal implied volatility in $[10^{-8}, 5]$;
- the solver is bisection, with at most 64 iterations;
- no price is ever clipped to a bracket, and no root is reported from outside it;
- a non-finite or negative price is rejected;

- a price within 10^{-10} of a European bound is rejected rather than inverted;
- the failure classes are named — invalid input, theoretical-bound violation, near-bound guard, outside bracket, numerical — and recorded per leg. The rejection of near-bound prices deserves a note. The classical approximation that the CPIV is the parity residual divided by the vega is first-order accurate in the no-arbitrage interior (Hentschel, 2003, analyzes the implied-volatility error that a price error induces directly); near the boundary vega approaches zero and the error amplifies without bound. Prices *on* the bound are therefore precisely those for which the local approximation is uninformative. The map is European because it is the map common to both legs, and because it is the map under which the synthetic truth is defined. Where the underlying option is American — the case for most listed equity options — the European map is an approximation for both legs, and the approximation error is common to the two legs to first order and largely cancels in the difference. Section 3.6 describes how the American exercise premium is handled.

3.3 The resolution rules

Certification is not a single test. Three rules must all be satisfied, and each is a rule about whether the *market* says something rather than about whether the *model* fits. **The time-value rule.** The out-of-the-money-equivalent time value of each leg is computed and compared with the declared resolution scale, one tick by default. The rule separates "the option is too cheap to say anything about" from "the option says something, and the inversion recovers it". **The finite-shock rule.** A declared positive shock and a declared negative shock are applied to the underlying, and the pair must produce valid roots under both. A pair producing a root under only one sign of shock has not been shown to be locally invertible, and is treated as unresolved. **The endpoint rule.** The CPIV implied by the four combinations of bid and ask endpoints is computed: the lowest from the call bid against the put ask, the highest from the call ask against the put bid. If an endpoint has no finite root, the range becomes one-sided and the pair becomes partially resolved. No endpoint is imputed and no missing endpoint is interpolated. The fourth rule is the tolerance rule, and it is what makes the abstention decision rather than merely a resolution decision. The pair is *certified* when the endpoint range of the CPIV is narrower than the declared tolerance, *identified but outside tolerance* when the range is finite but wider, and *unresolved* when the range cannot be formed. A pair identified outside the tolerance still has a number; the protocol declines to certify that it is determined to the standard asked for.

3.4 The status taxonomy and the reconciliation invariant

Every attempted pair ends in exactly one of seven terminal statuses: `RESOLVED_WITHIN_TOLERANCE`, `IDENTIFIED_OUTSIDE_TOLERANCE`, `PARTIALLY_RESOLVED`, `UNRESOLVED`, `INVALID_INPUT`, `TIMING_MISMATCH` and `CONTRACT_MISMATCH`. Reason codes are stored separately from the terminal status, so that a diagnostic can never stand in for a status. A terminal status that contradicts the stage at which the pair failed is a reconciliation error, and the run fails rather than continuing. The reconciliation invariant is that the terminal counts sum to the attempted count, and that each stage's count equals the sum of the terminal statuses downstream of it. A residual — a pair that entered the pipeline and left no trace — fails the run. The funnel reports, for every stage, the count, the share of attempted pairs, the share of the prior stage, and the dominant rejection reason. Dropping a pair reports coverage conditional on the survivors, and the survivorship depends on upstream decisions invisible in the final number; a terminal status for everything gives coverage an unambiguous denominator and a funnel that says where the pairs went.

3.5 Certification and abstention

A pair is certified when all three resolution rules are satisfied and the endpoint range of the CPIV is narrower than the tolerance. The certified set is what the protocol reports. The abstained set is everything else, and it is not empty by construction. In the development draw of the benchmark, 72 of 894 attempted pairs were certified, and 719 were identified but outside the tolerance, with the remainder split across partially resolved, invalid input, timing mismatch and contract mismatch. The abstention rate is not a cost to be minimised. It is the price of the claim being made, and the paper's secondary outcome (Section 6.9) puts a number on what that price buys in the one place a downstream user needs a *set* rather than a *number*: the calibration of a volatility surface.

3.6 Handling the American exercise premium

Most listed equity options are American, and the European inversion map ignores the early-exercise premium the put leg carries. If that premium is material, the two legs are being inverted under a model that is wrong for one of them, and the resulting CPIV is contaminated.

Two facts limit the damage. The premium is a *common* function of the underlying's parameters and the contract's moneyness and maturity, so it does not vary arbitrarily across the strike grid, and a surface fitted to a contaminated set is biased in a structured rather than a random way. And its magnitude is small relative to the uncertainty contributed by the quote spread, for most of the parameter range in which the protocol certifies anything at all.

The benchmark makes this explicit rather than assuming it away. The synthetic put leg is genuinely American, priced by an 800-step binomial lattice (Cox et al., 1979) rather than by a simulation-based American pricer (Longstaff & Schwartz, 2001), because the premium has to be exact for the truth to be exact. The call leg is worth exactly its European value because it carries no early-exercise premium without a dividend. The true CPIV is therefore non-zero by construction — it is the volatility-equivalent of the exercise premium — and the protocol's certified values recover it, with a median absolute error of 0.007 percentage points in the development draw (Section 6.2).

The alternative treatment, an American fallback for the put leg, is available and is a protocol change under the pre-registration rather than an extension of the results. It is not adopted here, because the licensed run has not fixed the exercise-style convention that would trigger it.

3.7 The calibration input set

The protocol's abstention removes pairs from any downstream use that needs a *set* of CPIV values, most obviously the calibration of a volatility surface. To make the cost of abstention concrete, one secondary outcome is declared in advance. At the primary tolerance, the paper counts how many contracts remain available to a surface calibration under the raw-midpoint input set and under the protocol input set, and reports how the fitted surfaces differ. The fit is a quadratic in log-moneyness, per expiry slice, on the CPIV surface. The comparison is descriptive: the paper does not claim that either surface is better specified, better behaved or more profitable. This outcome was adopted before the freeze as the bridge to the journal's computational focus: the one place where the protocol's behaviour has a consequence for a downstream numerical procedure.

4. A synthetic instrument with a known truth

4.1 What a synthetic market can say that a real one cannot

The advantage of the synthetic instrument is that the truth is defined before the quote exists, and the paper states what that means and what it does not. The generator draws, for each underlying and each trading day, a set of contracts at declared strikes and maturities. For each contract it computes the *unrounded fair value* of both legs, inverts those fair values under the same European map the protocol uses, and takes the difference as the true CPIV. Only after the truth has been fixed does it produce the observable quote: the fair value is rounded to a tick, a half-spread is added, the timestamp is jittered, and defects are injected at declared rates. The truth is therefore a property of the model, not of a market: it is the quantity a perfect measurement of this synthetic market would recover. The instrument can report *accuracy*, which a real market never can, and it can probe conditions that are hard to find in real data — wide quotes, coarse ticks, short maturities, a timing skew — because those can be dialled in exactly. What it cannot do is support a claim about any real security, index or option (Section 7.2).

4.2 The generator

The base specification, frozen in the pre-registration, is listed in Supplementary Appendix A. In outline: two underlyings at spot 100 and 250, 30 trading days each, maturities of 14, 30 and 60 days, five strikes from 90 % to 110 % of spot, a continuously compounded rate of 5 % with no dividend yield, a volatility surface with a term slope and a log-moneyness skew, a tick of 0.01 and a half-spread that widens in the wings, at the short end and on every seventh session, sub-second timestamp jitter, and injected defects at declared rates. This yields 894 attempted pairs in the development draw, of which 868 carried no defect: 11 stale quotes, 8 crossed quotes, 5 out-of-bounds quotes and 2 contract mismatches, one defect per remaining pair. The 6 contracts with a missing leg never become pairs and are not among the 894. The generator is a pure function of its seed and configuration: the same seed reproduces the same quotes, pair identifiers and truth values, which is what makes the replication design of Section 5.3 possible at all.

4.3 The truth

The true CPIV is negative for a structural reason: the American put is worth more than its European value, so the implied volatility recovered from it is *higher* than that recovered from the call, and the difference is negative. In the development draw the true spread ranges from -19.331 to -0.021 percentage points with a median of -0.248, and it is defined for all 894 attempted pairs. The magnitude matters. A true spread of a few tenths of a percentage point, against a quote-induced uncertainty of the same order, is the regime in which the measurement problem is genuinely hard: signal and noise are the same size. It is also the regime in which a protocol that reports a number without saying whether the number is determined is most likely to mislead.

4.4 What the instrument is not

The synthetic instrument is a *model artefact*, and three things follow. It does not license an external-validity claim. A protocol that behaves well on this instrument may behave differently on a real market, for the reasons in Section 2.6: coverage is determined by the microstructure of the market it meets, and the instrument's microstructure is one this study chose. It does not license a market claim of any kind; Section 7.2 lists the excluded claims. It

does not license a claim about the optimal tolerance: the specification sweep varies the tolerance on a synthetic market, which says nothing about which tolerance an analyst should use on a real one.

5. Design and pre-registration

5.1 The status of the pre-registration

The project's main pre-registration cannot be frozen, because the data gate is closed and the universe, the sampling rule and the tick convention all depend on a vendor's schema that does not yet exist. What *can* be frozen without a licence is everything about the protocol and the instrument, and that is what a separate addendum freezes: the generator specification, the protocol settings, the seed list, the specification grid, the nulls, the number of trials and the decision rules. The addendum was written before the replication sweep was run. The one base-market study that had already been run at that point is declared as the *development draw*, is reported in full for transparency, and is excluded from every tally in Section 6.6. The replication seeds were run after the addendum was written and before any of their outputs was read. The rules are evaluated by code, not by hand. `verdict.py` encodes the six rules and reads only the sweep outputs; the tally in Section 6.6 is its output. A code defect may be fixed after the freeze; a change to a rule, a seed, a specification or a threshold may not, since that would require a new addendum version, a fresh sweep and a new tally. This is the discipline the pre-registration literature argues for in its general form (White, 2000; Harvey et al., 2016; Nosek et al., 2018), applied at the level of a single study.

5.2 The declared settings

Every setting the protocol uses is a declared quantity, frozen in the addendum and listed in Supplementary Table S2: the inversion map, the root bracket, the near-bound guard, the resolution scale, the synchronisation tolerance, the tolerance grid and primary tolerance, the sign band, the perturbation set, the period split and the inference unit. No setting is a free parameter of the analysis.

The inference unit deserves a note. Options on the same underlying and day share a spot, a session volatility and a set of market-maker quotes, so treating pairs as independent would understate the uncertainty and would make the nulls of Section 2.5 too easy to beat. All intervals are therefore computed by resampling whole underlying-days, and the day-block null is the primary test.

5.3 The replication design

A single draw is one number, and one number cannot support a claim about a protocol. The design has two arms. The **seed sweep** repeats the entire study — a fresh market, a fresh protocol run, a fresh frontier — on 23 declared replication seeds, each a complete re-draw of the instrument. The seeds are a base seed plus a fixed stride, so the set is reproducible and cannot be adjusted after the fact. Each draw reports the same quantities, so the results are read as distributions rather than point estimates. The **specification sweep** varies the instrument and the protocol one factor at a time away from the base market. The market group varies volatility (low 15 %, high 35 %), quote width (two and three ticks), tick size (0.05), the wide-session period (removed), the strike grid (85 % to 115 %) and the maturities (7/14/30 days and 30/60/90 days). The protocol group varies the analyst's own knobs — the tolerance at 0.05, 0.10, 0.25 and 0.50 percentage points (sweep specifications, not the grid), and the resolution scale at 0.5, 1 and 2 ticks — which maps the coverage–error frontier an analyst

actually faces. The seed sweep asks whether the quantities are a property of the protocol or of one lucky draw; the market group, whether they are a property of the protocol or of one synthetic market.

5.4 The perturbation set

Six perturbations are declared, each a plausible accident of a real feed rather than an arbitrary stress: every quote widened by half a tick; the spot used for inversion shifted up by one basis point; the continuously compounded rate shifted up by five basis points; the underlying's timestamp moved three seconds behind the quotes; the resolution scale halved; and the synchronisation tolerance relaxed to five seconds. Table 7 reports the response of each, and Section 6.7 says what it means. The perturbations are diagnostics rather than confidence bounds. They answer "which parts of the result are load-bearing?", and the answer determines which components a reader should trust.

5.5 The decision rules

Six rules were declared, each a statement the benchmark can falsify.

Rule	Statement	Pass condition on the replication seeds
R1 — selection	The abstention rule selects a set that is genuinely easier to measure, not merely smaller.	Certified error below the day-block null mean in ≥ 16 of 23 draws, and day-block $p < 0.05$ in ≥ 16 of 23
R2 — own-set accuracy	The protocol's own-set error is lower than the raw midpoint's own-set error.	Own-set MAE(certified) < own-set MAE(raw) in ≥ 16 of 23 draws
R3 — coverage stability	The certification coverage does not depend on one particular synthetic market.	Across the 10 market specifications, the minimum certified coverage at the primary tolerance is ≥ 50 % of the median coverage
R4 — tolerance monotonicity	Coverage is non-decreasing in the certification tolerance.	Holds in ≥ 22 of 23 draws
R5 — period agreement	The primary and holdout periods agree.	Sign of the error reduction agrees in ≥ 16 of 23 draws, and holdout coverage is within 10 percentage points of primary coverage in ≥ 16 of 23
R6 — perturbation sanity	The perturbation set does not dominate the result.	No single perturbation changes certified coverage by more than 15 percentage points in absolute value

Two features of the rules are worth stating. The thresholds are counts of draws, not effect sizes, so the rules are about *consistency* rather than magnitude: a rule passes when the phenomenon is present in most independent re-draws. And R1 and R2 carry a declared consequence. If either fails, the benchmark shows that abstaining on hard pairs does not buy accuracy, and the paper's central mechanism claim must be reduced to coverage and stability only. That reduction would bind the abstract, the key messages and the conclusion, and could not be renegotiated afterwards.

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6. Results

6.1 The funnel

The funnel of the development draw (Table 1) is the first result, because it establishes what the protocol is actually rejecting.

Table 1. Attrition funnel, development draw (seed 20261022)

Stage	Count	Share of attempted	Share of prior stage	Dominant rejection reason
Contract	892	99.8 %	99.8 %	—
Timing	881	98.5 %	98.8 %	—
Quote	873	97.7 %	99.1 %	quote invalid
Input	873	97.7 %	100.0 %	—
Midpoint	868	97.1 %	99.4 %	theoretical bound violation
Endpoint	791	88.5 %	91.1 %	endpoint incomplete
Time value	791	88.5 %	100.0 %	—
Finite shock	791	88.5 %	100.0 %	—
Tolerance	791	88.5 %	100.0 %	—

Note: shares are percentages of the attempted total and of the preceding stage. The terminal statuses reconcile to the attempted total of 894 pairs.

The **endpoint** stage is where the largest single loss occurs, from 868 pairs to 791, and the dominant reason is an incomplete endpoint. This is the stage at which the market's own price granularity runs out: an endpoint combination that asks the inversion for a root on the far side of the no-arbitrage boundary has no root, and the protocol does not impute one. The **tolerance** stage removes nothing and separates what it receives: of the 791 resolvable pairs, 72 are certified at the primary tolerance and 719 are identified but not certified. Abstention is therefore 719 pairs held back by the tolerance, 77 lost to an incomplete endpoint, and 26 lost to a contract, timing or input defect: 822 of 894 attempted pairs, or 91.9 %. The terminal statuses reconcile as 72 resolved within tolerance, 719 identified outside tolerance, 77 partially resolved, 0 unresolved, 13 invalid input, 11 timing mismatch and 2 contract mismatch, summing to the 894 attempted pairs.

6.2 Accuracy, and the error profile

The protocol's advantage in the development draw is large, and it is important to be clear about what kind of advantage it is. The raw midpoint achieves a mean absolute error of 0.532 percentage points over 889 scored pairs; the protocol's certified subset achieves 0.009 over 72 pairs, a factor of 61 in the benchmark, computed from unrounded values. That number is a *selection* number. On the common set of 72 pairs the two methods are numerically identical and the difference in mean absolute error is exactly zero. The informative comparison is therefore not the one a reader will instinctively make, and Section 6.4 makes it. The error profile in Table 2 shows the shape the mean hides. The raw midpoint's median absolute error is 0.025 percentage points, so its mean of 0.532 is carried by a tail: a minority of pairs have errors an order of magnitude larger than the typical pair, and they are exactly the pairs a reader would most want to interrogate. The share of raw-midpoint pairs inside 0.10 percentage points is 82.1 %, against 100.0 % for the certified subset, whose maximum absolute error is 0.024. The tail is not noise smeared around an estimate; it is a set of pairs where the midpoint is simply not a measurement, and no downstream consumer can tell which those are. The identified set, at 791 pairs, sits between the two: its mean absolute error is 0.040, and its share inside 0.10 percentage points is 90.5 %. The protocol produces a graded statement — certified, identified but imprecise, unresolved — rather than a binary one, and the grading is informative.

Table 2. Error profile in the benchmark, development draw

Method	n	Mean abs. error	Median	p90	p95	Max	Within 0.10 pp
Raw midpoint	889	0.532	0.025	0.261	5.990	10.956	82.1 %
Bound-filtered	889	0.532	0.025	0.261	5.990	10.956	82.1 %
Identified	791	0.040	0.021	0.098	0.143	0.415	90.5 %
Certified	72	0.009	0.007	0.018	0.021	0.024	100.0 %

Note: all errors are absolute errors in percentage points of volatility against the benchmark's defined truth, on each method's own set. The bound-filtered baseline is numerically identical to the raw midpoint in this draw, because the bound filter removed no pair the raw midpoint could otherwise have used.

6.3 Coverage

The coverage of certification in the development draw is 8.1 % of attempted pairs at the primary tolerance of 0.10 percentage points, which is 72 pairs. At the tighter tolerance of 0.05 percentage points the coverage is 0.0 % and no pair at all is certified; at the looser tolerance of 0.25 percentage points it is 32.8 %, or 293 pairs.

A coverage of zero at the tightest tolerance is not a degenerate case: the benchmark's quote widths and its true spread are of comparable size, so the 0.05 pp standard is beyond what the quotes support for essentially every pair. A practitioner who demands a tighter standard than the market supports should receive nothing, and does.

The error avoided by abstention — the raw-midpoint error on the pairs the protocol declined to certify — is 0.578 percentage points in the development draw, with interval [0.547, 0.606]. This is the price of the abstention expressed as an error rather than a coverage: the protocol declines to certify the pairs that would have been wrong by close to six tenths of a percentage point on average.

6.4 The selection frontier

The frontier (Table 3) is the paper's central result, because it is the only result that speaks to the question the protocol poses. At the primary tolerance of 0.10 percentage points, the certified subset in the development draw has a mean absolute error of 0.009 percentage points, while a pair-null draw of the same size has a mean of 0.524 and a day-block null a mean of 0.534. The certified set is far below both nulls, and the add-one p-values are 0.0005 against the pair null and 0.0005 against the day-block null.

Table 3. The selection frontier in the benchmark, development draw

Tolerance (pp)	Certified	Coverage	MAE certified	MAE abstained	Pair null mean	Block null mean	p (pair)	p (block)
0.075	6	0.7 %	0.008	0.535	0.542	0.531	0.0035	0.0045
0.10	72	8.1 %	0.009	0.578	0.524	0.534	0.0005	0.0005
0.15	157	17.7 %	0.010	0.643	0.530	0.532	0.0005	0.0005
0.25	293	33.0 %	0.014	0.786	0.534	0.530	0.0005	0.0005

Note: each null draws a subset of exactly the certified size from the same scored universe, over 2000 Monte Carlo trials. The day-block null draws whole underlying-days with replacement. The p-values are add-one estimates. The 0.05 pp row is omitted because no pair is certified at that tolerance in this draw. Coverage is the scored-universe share; Section 6.3

reports the share of attempted. The rows are grid points plus two intermediate points of the seven-point axis in Supplementary B.3.

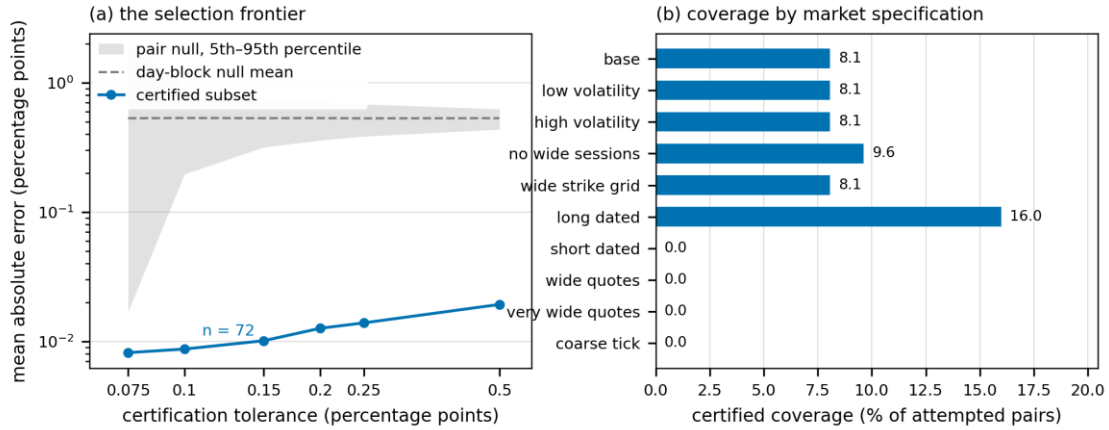


Figure 1. *(a) Certified mean absolute error against the certification tolerance, with the pair-null 5th to 95th percentile band and the day-block null mean, on a log scale; the nulls sit three orders of magnitude above the certified error. (b) Certified coverage across the ten market specifications at the primary tolerance. Benchmark only; development draw throughout.*

Two readings are possible. The favourable one: the rule is picking out pairs that are genuinely easier to measure, which a random selection would not. The sceptical one: the certified subset is small, and a small subset of a noisy population tends to have a smaller error even when the selection is random. The size-matched nulls address the second reading, and the certified set beats both by a wide margin — in this draw. Section 6.6 reports the same test across 23.

6.5 Replication across draws

The seed sweep repeats the whole study on 23 declared replication seeds. The headline quantities are reported as distributions (Table 4).

Table 4. The replication distribution, 23 draws in the benchmark

Quantity	Median	IQR	Range
Certified coverage	8.2 %	8.1 – 8.5 %	7.9 – 10.1 %
Certified MAE	0.009 pp	0.008 – 0.010	0.008 – 0.010
Raw midpoint MAE	0.538 pp	0.530 – 0.542	0.516 – 0.550

Note: each draw is a complete re-draw of the instrument and a complete re-run of the protocol. The development draw is excluded from this table, as the pre-registration requires.

Median certified coverage is 8.2 %, with a full range of 7.9 % to 10.1 % across the draws: a stable yield in the sense that matters, but a low one. The day-block null p-value has a maximum of 0.0005, which is the floor of the test — with 2000 trials the add-one estimator cannot report less than 0.0005 — so the reading is that the selection effect is outside the null in every draw, not that its tail probability is known to five decimal places.

6.6 The pre-registered tally, including the failure

The tally (Table 5) is the output of `verdict.py`, which applies the rules of the pre-registration addendum to the sweep outputs and to nothing else. The development draw is excluded.

Table 5. The pre-registered rule tally, 23 replication draws

Rule	Statement	Outcome
R1	The abstention rule selects a set that is genuinely easier to measure, not merely smaller	PASS
R2	The protocol's own-set error is lower than the raw midpoint's	PASS
R3	Certification coverage does not depend on one particular synthetic market	FAIL
R4	Coverage is non-decreasing in the certification tolerance	PASS
R5	The primary and holdout periods agree	PASS
R6	The perturbation set does not dominate the result	PASS

R1 passed. The certified set's mean absolute error was below the day-block null mean in 23 of 23 draws, and the day-block null p-value was below 0.05 in 23 of them; the rule required 16. That is the substantive result: against the null that respects the within-day dependence of the quotes, the certified set is better, not merely smaller. **R2 passed too**, and it is the weaker, partly redundant companion: the certified set's own-set error was below the raw midpoint's in 23 of 23 draws, against a required 16. **R3 failed.** This is the paper's most important negative result and it is reported as prominently as the passes. The rule required the minimum certified coverage across the ten market specifications to be at least half the median coverage. The observed ratio is 0.000, with a minimum coverage of 0.0 % against a median of 8.1 %.

Table 6. Certified coverage across the market specifications in the benchmark

Specification	Certified coverage	Certified MAE (pp)	Raw midpoint MAE (pp)
base	8.1 %	0.009	0.532
low_vol	8.1 %	0.008	0.441
high_vol	8.1 %	0.009	0.038
wide_quotes	0.0 %	—	0.622
very_wide_quotes	0.0 %	—	0.710
coarse_tick	0.0 %	—	1.161
no_wide_sessions	9.6 %	0.008	0.512
wide_strike_grid	8.1 %	0.010	0.731
short_dated	0.0 %	—	1.771
long_dated	16.0 %	0.008	0.037

The pattern in Table 6, and in the right-hand panel of Figure 1, is not noise. The specifications that widen the quotes or coarsen the tick drive coverage towards zero, because the endpoint range of the CPIV widens with the quote width and the resolution rule abstains on options whose time value has fallen below the scale of the instrument. A protocol whose coverage depends this strongly on the microstructure is not a portable rule with a fixed yield, and a reader who carries the development draw's 8.1 % coverage figure into a different market will be wrong, possibly by a factor of several. The honest reading is that this is a *property of the measurement*. A protocol that certified everything would be certifying pairs whose CPIV is not determined by the quotes, and one that abstained everywhere would be useless. The protocol behaves correctly in both extremes; what it cannot do is promise a yield in advance, and the R3 failure is the evidence for that statement. **R4 passed.** Coverage was monotone in the tolerance in 23 of 23 draws, against a requirement of 22. A weak condition, but the one

that would detect a coding error in the tolerance rule. **R5 passed.** The sign of the own-set error reduction agreed between the primary and holdout periods in 23 of 23 draws, and the two periods' certified coverage were within 10 percentage points in 23 of them, against a required 16 in both cases. The period split is a weak test, since the holdout was not used for threshold selection and the two periods share the same generator, but it does show the result does not depend on the first half of the sample. **R6 passed.** The largest absolute change in certified coverage caused by any single perturbation was 8.1 percentage points, against a threshold of 15. Section 6.7 discusses the perturbation response.

6.7 The perturbation response

Table 7. Perturbation response in the benchmark, development draw

Perturbation	Coverage change	MAE (pp)	Mean abs. CPIV change (pp)
Endpoint widened by half a tick	-7.3 pp	0.016	0.000
Spot +1 bp	0.0 pp	0.066	0.066
Rate +5 bp	0.0 pp	0.054	0.054
Spot timestamp −3 s	-8.1 pp	—	—
Resolution scale halved	0.0 pp	0.009	0.000
Synchronisation tolerance relaxed to 5 s	0.0 pp	0.009	0.000

The perturbations divide cleanly into three groups, and the division is informative. The **timing** perturbation is the largest in coverage terms. Moving the spot's timestamp three seconds behind the quote timestamps exceeds the synchronisation tolerance, certified coverage falls to zero, and the protocol reports no measurement at all. This is correct behaviour and worth pausing on: a stale underlying price, a completely routine feed condition, does not merely add noise to a CPIV, it destroys the measurement. A pipeline that does not check timestamps will still report numbers. Widening every quote by half a tick does comparable but smaller damage: certified coverage moves by 7.3 percentage points, because the endpoint range widens and the tolerance rule tightens. The **model-input** perturbations, a one-basis-point spot shift and a five-basis-point rate shift, leave the coverage untouched but move the CPIV values: the mean absolute change is 0.066 and 0.054 percentage points respectively — 8 and 6 times the certified subset's own error of 0.009. This is an important and slightly uncomfortable finding: the protocol certifies a pair as determined with respect to the *quote* uncertainty and is silent about the *model-input* uncertainty, which in these units is larger. The certification is a statement about the quotes, not about the inputs. The **protocol-setting** perturbations, by contrast, change nothing here: the resolution scale is not binding at the base market's microstructure, consistent with the endpoint stage, not the time-value stage, being where pairs are lost.

6.8 Strata

Certification is not spread uniformly across the contract space. In the development draw it is confined to the 60-day maturity, to pairs quoted at a single-tick half-spread, and to the three central strikes; it is zero in every wing bucket, at 14 and 30 days, at half-spreads of two ticks or more, and among true spreads below 0.1 percentage points. Table S1 in the supplementary material reports the full cross-tabulation, with the raw midpoint's error in each cell for comparison. The strata are also the result most likely to be over-read. They are computed *in the benchmark*, against a truth defined by the benchmark's model. A certified set concentrated near the money is evidence that the protocol's criteria behave as intended; it is not evidence

about where certification would concentrate in a real market, which depends on that market's own smile, spreads and liquidity calendar.

6.9 The secondary outcome: what abstention costs a surface calibration

The secondary outcome is the calibration of a volatility surface, which needs a *set* of CPIV values rather than a single one. In the development draw, the raw-midpoint input set contains 296 contracts and the protocol's input set contains 72, so the calibration retains 24.3 % of the raw set and forgoes 224 contracts. Fitting a quadratic in log-moneyness per expiry slice, the two surfaces differ by at most 2.227 percentage points of CPIV, two orders of magnitude larger than the certified subset's own measurement error of 0.009. The comparison is descriptive only: the paper makes no claim that either surface is better specified, better behaved or more profitable. The surface difference is the quantity to keep in view. Abstaining on 75.7 % of the contracts moves the fitted surface by more than the measurement error of either set, and a practitioner who silently substituted one for the other would not know it. Abstention is not free in calibration, where the unit is a set; for one pair the calculus reverses (Section 6.2).

7. Discussion

7.1 What the benchmark establishes

Three things are established in the benchmark, and a fourth is not. **The abstention rule selects a set that is genuinely easier to measure.** This is R1, and it passed in 23 of 23 replication draws against the conservative null: the certified pairs have a lower mean absolute error than a size-matched random subset of the same scored universe, even when that subset preserves the within-day dependence of the quotes. The rule is not selecting fewer pairs, and it is not selecting a lucky day. **The protocol's own-set advantage is a selection effect, and the paper says so.** The common-set identity is reported as exactly zero rather than as a small number, and the frontier p-values sit beside every own-set comparison. The distinction between "this method is better" and "this method selected better data" is the difference between a reproducible method and a result that depends on the analyst's discretion. **The thresholds can be fixed before any market data is read.** The addendum froze the generator, the settings, the seeds, the specifications, the nulls and the rules before the replication outputs were read. **What is not established is any market claim.** The benchmark contains no market data and supports no statement about any real security, index or option.

7.2 The non-claims

The following claims are excluded by the design and bind every part of this paper, including the abstract and the conclusion: any estimate of coverage, error or stability in a real market; any statement about a real security, index or option; any claim of trading profitability, alpha or forecasting value; any claim that the certification tolerance is universal or optimal; and any external-validity claim from the synthetic market to real quotes. Wherever a number appears in this paper, it is a number *in the benchmark*. The largest difference between the benchmark and a licensed run is that the benchmark's truth is known, and a known truth makes accuracy measurable in a way that no real market ever permits. A protocol validated on accuracy and then applied to a market where accuracy is unobservable has inherited its thresholds but none of its evidence about them.

7.3 Threats to validity

The truth is a model artefact, and the European map is an approximation for American options. The benchmark's put leg is American and is inverted under a European map, and the protocol recovers the truth anyway — but the cancellation that makes this work depends on the exercise premium being a smooth function of the contract's parameters. A market in which early exercise is discontinuous in the strike would break that, and the licensed run may present departures the benchmark does not. **The instrument's microstructure was chosen.** The R3 failure is a statement about the sensitivity of coverage to microstructure, demonstrated on specifications of *this* instrument. The qualitative finding — that coverage is a property of the market, not of the protocol — is robust to this, but the numbers in Table 6 are not. **The inference is on one instrument with two underlyings.** The underlying-day block bootstrap treats the underlying-day as the unit, and there are 60 such blocks in the development draw. The replication across seeds is the paper's defence against the precision of any one interval. The primary and holdout periods share the generator and the underlying paths, so the split tests robustness rather than replication.

7.4 What a practitioner should do with this

The operational message is deliberately narrow. A practitioner who wants to use a CPIV should know three things. First, a CPIV computed from midpoints is not a measurement, and the benchmark's raw midpoint has a mean absolute error of 0.532 percentage points with a 95th percentile of 5.990 and a maximum of 10.956. Those are benchmark numbers and transfer to a real market only as an order of magnitude, but the shape they describe — a small tail carrying a large mean — is generic. Second, the errors concentrate in identifiable places: wide quotes, short maturities, wings, and the ticks on the clock. A pipeline that checks those four things will abstain on most of what it should, and that is cheaper and more robust than a full abstention protocol. Third, and least obvious: model-input sensitivity is larger than quote sensitivity in these units. A certified CPIV is a statement about the quotes, not about the spot, the rate or the dividend assumption behind them; a user who reads "certified" as "correct" has misread it (Section 6.7).

8. Conclusion and future research

A call–put implied-volatility spread computed from quote midpoints is routinely reported as though it were a measurement, and it frequently is not. This paper describes a protocol that decides, for each quote pair, whether the spread is determined at a declared tolerance, declines to assert when it is not, and assigns every attempted pair one terminal status so that nothing disappears silently. Because the protocol labels rather than re-estimates, its advantage over the raw midpoint is a selection effect, and the paper reports it as one: the common-set difference is exactly zero, and the informative comparison is against size-matched random subsets. Against a null that resamples whole underlying-days, and so respects the within-day dependence of the quotes, the certified subset is easier to measure in 23 of 23 replication draws in the benchmark. One pre-registered rule failed, and it is the result with the longest reach: certification coverage depends strongly on quote microstructure, falling from 8.1 % at the median market specification to 0.0 % at the least favourable one. The protocol is not a portable rule with a fixed yield. It is a labelling device whose coverage is a property of the market it meets, and a practitioner who carries a coverage figure from one market into another will be wrong. The benchmark supports no market claim, and none is made. What it does establish is that the machinery behaves as specified, that the abstention rule has a quantifiable selection effect, and that the thresholds can be fixed before any market data is read. The

research that follows is of three kinds. **A licensed run**, whose schema, sampling rule and tick convention must be fixed and frozen first, and whose primary result will be coverage and stability rather than accuracy, because no real market permits accuracy to be measured. **An American fallback for the put leg**, which the benchmark shows the European map can survive in the tested regime, and which must be adopted on the strength of the licensed data's exercise-style convention rather than a synthetic one. And **an independent reimplementation** of the inversion and the resolution rules, which no single-authored pipeline can supply for itself.

Declarations of Interest

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Competing interests. The author reports no conflicts of interest. The author alone is responsible for the content and writing of the paper.

Ethics approval. This study uses a synthetic instrument defined by a released generator, with no human participants, human tissue, personally identifiable data, or data collected from persons. Ethical review and informed consent were therefore not required, and no institutional review board approval applies.

Data and code availability

This paper contains **no market data**. Every result comes from a synthetic instrument whose true call–put implied-volatility spread is defined by the generator before any quote exists. The instrument, the protocol, the replication sweep, the decision-rule tally and the manuscript build are implemented in code, and run manifests record the configuration, the seed, the environment and the output hashes of every run reported here. The code and the aggregate run artefacts (the study summary, the per-draw sweep outputs, the specification grid and the rule tally) are available from the author for review and will be deposited in a public repository on acceptance.

No licensed market data was used. The licensed run on which the protocol is intended to be applied has not been carried out, and no result in this paper depends on one.

Use of artificial intelligence

The author used Perplexity for deep research. An AI assistant helped draft the phase-0 design record and the pre-registration, wrote and maintained the measurement and analysis code and its test suite, the build, rendering and submission scripts and the manuscript-limits checker, and assisted with the language and structure of this manuscript. No AI system generated a result, a number or a table: every number in this paper is produced by the released code from the released artefacts, and the manuscript build fails if any numeric slot in the text has no corresponding value in those artefacts. The author reviewed all assisted work and is responsible for the entire content of the paper.

Author contributions

The sole author contributed to every CRediT category: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization, and Supervision. Author identity is withheld for

blind review; the named author appears on the separate title page. The author reviewed and approved every AI-assisted task described above and takes full responsibility for the paper.

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